22 Show that $\frac{6x^3}{(9x^2 - 144)} \div \frac{2x^4}{3(x - 4)}$ can be written in the form $\frac{1}{x(x + r)}$ where r is an integer.

Invert the second fraction and multiply:

$6x^3 / (9x^2-144) \times 3(x-4) / (2x^4)$

$9x^2-144 = 9(x-4)(x+4)$

Cancel common factors to get $1/[x(x+4)]$.

Thus $r = 4$.

1 mark
invert and multiply

1 mark
factorise denominator

1 mark
r=4

(Total for Question 22 is 3 marks)

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